Homework #3: Kalman Filter Implementation for GPS and IMU Fusion

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October 6, 2025

1 Introduction

This report presents a Kalman Filter implementation for fusing GPS and IMU data to estimate vehicle pose in 2D. The basic challenge is straightforward: IMU sensors give us high-frequency measurements but drift over time, while GPS provides accurate global positions but only updates once per second. We need both to work reliably.

2 Kalman Filter Equations

2.1 State Vector

The state vector is defined as:

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ \theta_{gps} \\ v_x \\ v_y \\ \theta_{integrated} \\ \text{heading_offset} \end{bmatrix} \tag{1}$$

where x, y are vehicle position in ENU coordinates, θ_{gps} is GPS-derived heading, v_x, v_y are velocity components, $\theta_{integrated}$ is continuously integrated gyroscope heading, and heading_offset is the calibrated offset between GPS and integrated heading.

2.2 Process Model

The process model describes how the state evolves over time:

$$\mathbf{x}_{k+1} = \mathbf{F}\mathbf{x}_k + \mathbf{B}\mathbf{u}_k + \mathbf{w}_k \tag{2}$$

The state transition matrix $\mathbf{F}$ is:

$$\mathbf{F} = \begin{bmatrix} 1 & 0 & 0 & \Delta t \cdot \text{vel_scale} & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & \Delta t \cdot \text{vel_scale} & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \tag{3}$$

The heading determination logic is:

$$\theta_{use} = \begin{cases} \theta_{gps} & \text{if GPS very recent (t ¡ 0.5s)} \\ \theta_{integrated} + \text{heading_offset} & \text{if GPS unavailable and calibrated} \\ \theta_{integrated} & \text{otherwise} \end{cases} \tag{4}$$

Body-frame accelerations are rotated to world coordinates:

$$a_{x,world} = a_{x,body} \cos(\theta_{use}) - a_{y,body} \sin(\theta_{use}) \tag{5}$$

$$a_{y,world} = a_{x,body} \sin(\theta_{use}) + a_{y,body} \cos(\theta_{use}) \tag{6}$$

Velocity and position updates:

$$v_{x,k+1} = v_{x,k} + a_{x,world} \cdot \Delta t \cdot \text{acc_weight} \tag{7}$$

$$v_{y,k+1} = v_{y,k} + a_{y,world} \cdot \Delta t \cdot \text{acc_weight} \tag{8}$$

$$x_{k+1} = x_k + v_{x,k+1} \cdot \Delta t \cdot \text{vel_scale} \tag{9}$$

$$y_{k+1} = y_k + v_{y,k+1} \cdot \Delta t \cdot \text{vel_scale} \tag{10}$$

2.3 Measurement Model

The measurement model relates GPS position measurements to the state:

$$\mathbf{z}_k = \mathbf{H}\mathbf{x}_k + \mathbf{v}_k \tag{11}$$

The measurement matrix $\mathbf{H}$ simply extracts the position components from the state vector, since GPS directly provides x and y coordinates. This direct relationship makes the measurement model straightforward and computationally efficient:

$$\mathbf{H} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \tag{12}$$

2.4 Covariance Matrices

2.4.1 Process Noise Covariance (Q)

The process noise covariance matrix models uncertainty in the state prediction process. Different state components have varying noise levels based on their physical characteristics:

$$\mathbf{Q} = \text{diag}([Q_{val}, Q_{val}, Q_{val} \cdot 0.1, Q_{val} \cdot 5, Q_{val} \cdot 5, Q_{val} \cdot 0.5, Q_{val} \cdot 0.01]) \tag{13}$$

where $Q_{val} = 0.005$ is the base process noise value. The scaling factors reflect the relative uncertainty of each state component.

2.4.2 Measurement Noise Covariance (R)

The measurement noise covariance adapts based on GPS signal quality to reflect the varying accuracy of GPS measurements:

$$\mathbf{R} = \begin{cases} \text{diag}([R_{good}, R_{good}]) & \text{if GPS status} = 0 \text{ (good)} \\ \text{diag}([R_{poor}, R_{poor}]) & \text{if GPS status} = 1 \text{ (poor)} \end{cases} \tag{14}$$

where $R_{good} = 4.0$ and $R_{poor} = 16.0$. This adaptive approach allows the filter to trust high-quality GPS measurements more than degraded ones.

2.4.3 Initial State Covariance (P)

The initial state covariance matrix reflects our initial uncertainty in the state estimate, with higher values indicating greater initial uncertainty:

$$\mathbf{P}_0 = \text{diag}([5, 5, 0.5, 2, 2, 0.5, 0.5]) \tag{15}$$

3 Initial Conditions

3.1 State Initialization

We start with the first valid GPS measurement for position: $x_0 = \text{GPS}_x[0]$ and $y_0 = \text{GPS}_y[0]$. The initial heading $\theta_{gps,0}$ comes from the GPS velocity vector. Both velocity components $v_{x,0}$ and $v_{y,0}$ start at zero since the vehicle begins stationary. The integrated heading $\theta_{integrated,0}$ initially matches the GPS heading, and the heading offset starts at zero before any calibration happens.

3.2 IMU Bias Estimation

The first five seconds of stationary data give us the IMU biases. Averaging accelerometer readings during this period yields biases of -0.518 m/s^2 (x), 0.0049 m/s^2 (y), and -9.532 m/s^2 (z). The gyroscope biases work the same way: -0.0016 rad/s, -0.0074 rad/s, and -0.0363 rad/s for x, y, and z axes.

4 Results and Analysis

4.1 Trajectory Comparison

Processing 535,754 IMU measurements and 21,871 GPS measurements, the filter produces smooth trajectory estimates that clean up GPS noise without losing accuracy. Running at 100 Hz matches the IMU rate, giving continuous pose estimates between the 1 Hz GPS updates. The heading calibration converges to an offset of -82.40 when GPS is present, and the filter switches between GPS and calibrated IMU heading as needed.

4.2 Translational Component Analysis

GPS works well when satellites are visible—the measurements are reliable and give stable position estimates for most of the route. Then the vehicle enters tunnels. GPS drops completely, and we're stuck with pure IMU dead reckoning. This is where things fall apart.

The position estimates degrade badly in tunnels. The problem isn't really the accelerometers—it's the heading. When heading estimation goes wrong, the coordinate transformation of accelerometer data goes wrong too. Small heading errors don't stay small. They accumulate fast during integration, causing the estimated trajectory to drift sideways from the true path.

When GPS is available, the filter weights these reliable measurements appropriately and maintains accuracy. In tunnels, it switches to dead reckoning with the calibrated heading offset, but this only helps so much. Dead reckoning quality lives or dies by heading accuracy. After exiting a tunnel and reacquiring GPS, the filter snaps back to the correct trajectory quickly. Without these periodic GPS corrections, long-term accuracy just isn't possible.

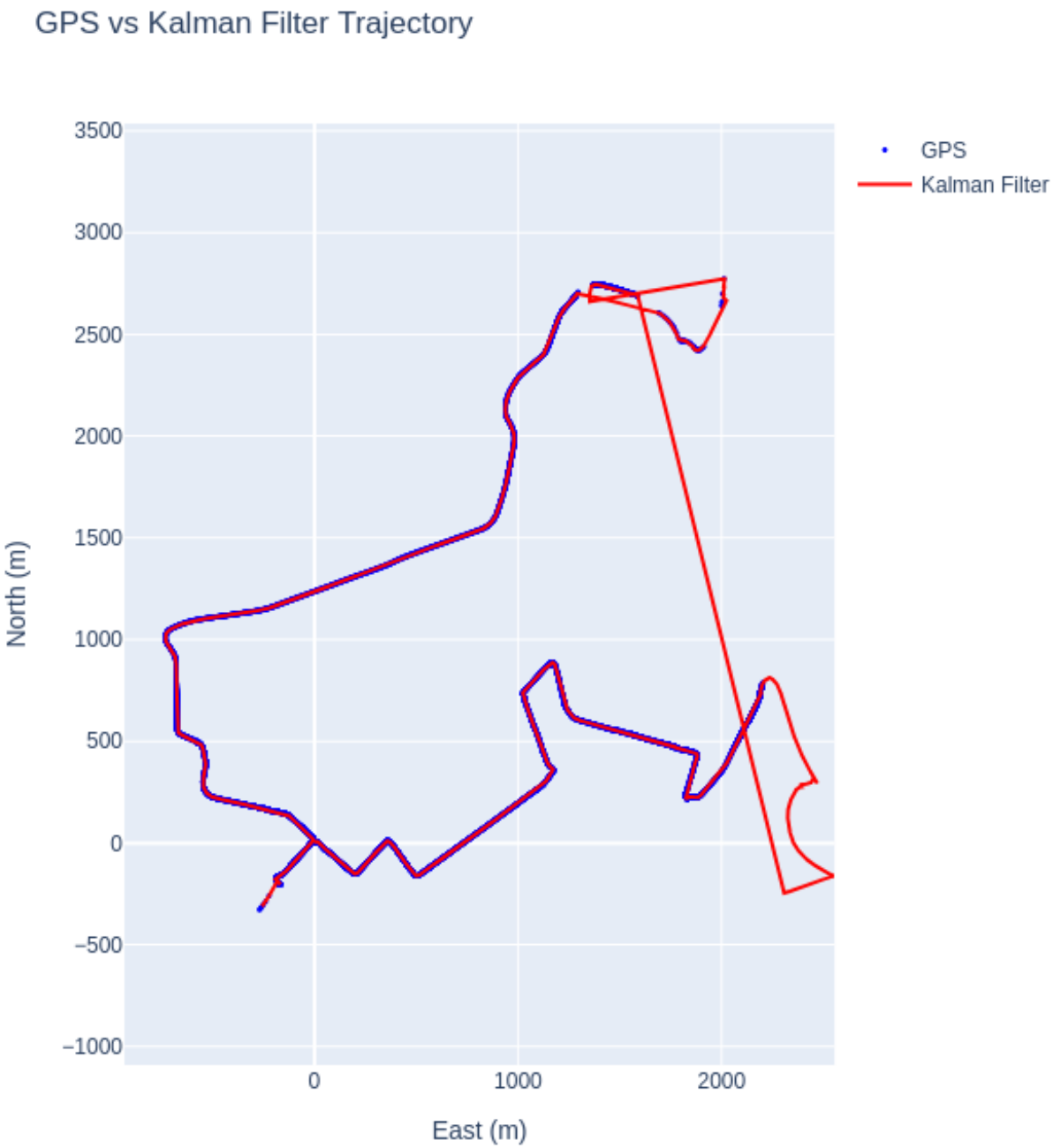

Figure 1: Enter Caption

4.3 Rotational Component Analysis

Heading is the real problem here. With GPS available, we calculate heading from velocity vectors—reliable, accurate, perfect for calibrating the IMU offset. The filter finds a −82.40

offset and applies it to the integrated gyroscope measurements.

But in tunnels, GPS disappears and we're left with pure gyroscope integration. The yaw data is very unreliable and noisy. Significant bias drift and noise cause the integrated heading to diverge quickly from reality which is the main bottleneck for dead reckoning performance. Heading errors get amplified when transforming accelerometer measurements, turning small orientation mistakes into large position errors.

Poor yaw quality during GPS outages makes dead reckoning nearly useless in tunnel sections. Heading errors compound continuously. Without GPS corrections, there's no way to maintain accurate orientation. Exiting a tunnel and reacquiring GPS means correcting for whatever heading error accumulated—sometimes quite large after extended blackouts. This is just a fundamental limitation of low-cost MEMS IMUs. You need accurate heading for dead reckoning to work, and these sensors simply can't deliver it without external references.

In the plots below, the middle part where blue data points are missing corresponds to the long tunnel where the GPS signal is lost:

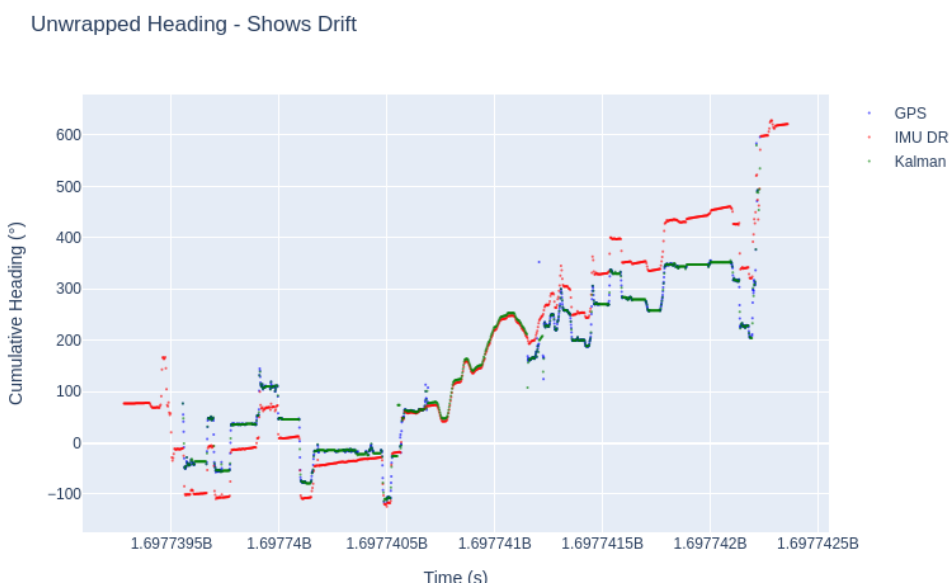

Figure 2: Unwrapped rotational components.

Figure 3: Wrapped rotational components.

5 Advantages and Limitations

5.1 Advantages

Fusing GPS and IMU through the Kalman filter significantly improves position estimation. GPS noise gets filtered out while preserving accuracy. Running at 100 Hz fills the gaps between 1 Hz GPS updates, giving continuous position estimates. With GPS present, the fusion approach uses reliable measurements to prevent drift and calibrate the IMU heading offset for later use. Adaptive noise modeling handles varying GPS quality well—poor signals don't corrupt the estimate, while good signals get their deserved weight.

5.2 Limitations

Position estimation quality collapses during GPS outages. Poor gyroscope quality is the culprit. Heading errors directly corrupt the coordinate transformation of accelerometer data, causing rapid lateral drift in tunnels. The filter can't fix what isn't fixable: poor yaw data means poor dead reckoning, period. Position errors accumulate without bound during long GPS blackouts because the IMU alone can't maintain heading accuracy. Only GPS reacquisition restores accuracy, which severely limits the practical value of dead reckoning through extended tunnel sections.

6 Discussion

The Kalman filter successfully addresses the sensor fusion challenge by combining the complementary strengths of GPS and IMU sensors. GPS provides globally accurate but low-

frequency position measurements, while the IMU offers high-frequency motion measurements that drift over time. The adaptive calibration strategy continuously estimates and corrects the IMU heading offset using GPS measurements, enabling reasonably robust operation during short GPS outages by maintaining an accurate reference frame.

The results show significant improvement over raw GPS measurements in several key areas. The filtered trajectory is much smoother by removing GPS noise while keeping accuracy. The filter provides pose estimates at 100 Hz compared to GPS updates at 1 Hz, enabling responsive real-time navigation. The system maintains reasonable performance across varying GPS signal quality through adaptive noise modeling that properly weights measurements based on their reliability.

7 Conclusion

The Kalman filter implementation successfully fuses GPS and IMU data to achieve robust 2D pose estimation for vehicle navigation. The adaptive calibration strategy effectively compensates for IMU drift while maintaining high-frequency state updates, demonstrating how sensor fusion can improve navigation accuracy and reliability for autonomous vehicle applications. The results show clear advantages in trajectory smoothness, drift compensation, and high-frequency pose estimation compared to using either sensor independently.

The implementation highlights the importance of initial calibration and careful parameter tuning for optimal performance, while also revealing limitations in GPS-denied environments and during extended outages. These findings provide valuable insights for developing more sophisticated navigation systems and suggest directions for future work, including extension to 6-DOF estimation, adding more sensor types, and developing more robust initialization procedures that don't require initial GPS availability.